Question 4: Install and Configure Git and Jenkins on Ubuntu Linux

Course: DevOps Foundations & Continuous Integration (Lab 1)
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Weightage: 5 Marks
Submission Component: Installation Steps, Verification Commands, Global Configuration, and System Administration

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1. System Environment Prerequisites

- **Target OS:** Ubuntu Linux 22.04 LTS (Jammy Jellyfish) / 24.04 LTS
- **Architecture:** x86_64 / amd64
- **Privileges:** Sudo / Root administrative access
- **Network Access:** Outbound HTTPS for package repositories and GitHub integration

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2. Installing and Configuring Git on Ubuntu Linux

Step 1: Update System Package Index

Before installing packages, synchronize the local APT package index:

```
sudo apt update
```

Step 2: Install Git via APT Package Manager

```
sudo apt install -y git
```

Step 3: Verify Git Binary & Version

Verify that Git is correctly installed in system PATH (`/usr/bin/git`):

```
git --version
# Expected Output: git version 2.34.1 (or latest available)
which git
# Expected Output: /usr/bin/git
```

Step 4: Configure Global Git Identity

Set user identity across all repositories on the workstation:

```
# Set user display name
git config --global user.name "Akash"

# Set user email address
git config --global user.email "ad1646414@gmail.com"
```

```
# Set standard initial branch name to main
git config --global init.defaultBranch main

# Set preferred text editor
git config --global core.editor nano

# Set safe line-ending behavior
git config --global core.autocrlf input
```

Step 5: Verify Global Git Configuration

Inspect active settings stored in `~/.gitconfig`:

```
git config --list --show-origin
# Output:
# file:/home/ubuntu/.gitconfig   user.name=Akash
# file:/home/ubuntu/.gitconfig   user.email=ad1646414@gmail.com
# file:/home/ubuntu/.gitconfig   init.defaultbranch=main
# file:/home/ubuntu/.gitconfig   core.editor=nano
# file:/home/ubuntu/.gitconfig   core.autocrlf=input
```

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3. Installing and Configuring Jenkins on Ubuntu Linux

Step 1: Install Java Runtime Environment (OpenJDK 17)

Jenkins is built on Java and requires OpenJDK 17 or 21:

```
sudo apt install -y openjdk-17-jdk
java -version
# Expected Output:
# openjdk version "17.0.10" 2024-01-16
# OpenJDK Runtime Environment (build 17.0.10+7-Ubuntu-122.04.1)
# OpenJDK 64-Bit Server VM (build 17.0.10+7-Ubuntu-122.04.1, mixed mode, sharing)
```

Step 2: Add Official Jenkins Repository Keyring

Modern Ubuntu releases require GPG keys stored in `/usr/share/keyrings/`:

```
curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | \\
  sudo tee /usr/share/keyrings/jenkins-keyring.asc > /dev/null
```

Step 3: Add Jenkins Debian Repository to APT Sources

```
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/" | \\
  sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null
```

Step 4: Update Index & Install Jenkins

```
sudo apt update
sudo apt install -y jenkins
```

Step 5: Enable & Start Jenkins Systemd Service

```
sudo systemctl enable jenkins
sudo systemctl start jenkins
```

```
sudo systemctl status jenkins --no-pager
# Output indicates: Active: active (running)
```

Step 6: Configure Firewall for Jenkins (Port 8080)

```
sudo ufw allow 8080/tcp
sudo ufw reload
```

Step 7: Retrieve Initial Admin Unlock Password

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
# Example Output: a8f49e0b1c234a5d89ef412356bc7890
```

Step 8: Web Dashboard Setup Wizard

1. Open browser to `http://<server-ip>:8080`.
2. Paste the initial admin password.
3. Select **"Install suggested plugins"** (Git, Pipeline, GitHub, Credentials).
4. Create primary Administrator user profile (`Akashraj`).
5. Confirm Jenkins Root URL (`http://<server-ip>:8080`).

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4. Verification & Audit Checklist

Component	Verification Command	Expected Status
Git Binary	`git --version`	`git version 2.x`
Git User Name	`git config --global user.name`	`Akash`
Git User Email	`git config --global user.email`	`ad1646414@gmail.com`
Java JDK	`java -version`	`openjdk version "17.x"`
Jenkins Service	`sudo systemctl is-active jenkins`	`active`
Port 8080 Binding	`ss -tulpn | grep 8080`	`LISTEN` on `:::8080`